Your Section Title

An optional subtitle or tagline

10:30 AM

Lectures

cats

demo.exe

Proof Assistant Demo

A gentle introduction to Lean

J. Appleseed --- Fall 2026

OK

README

Exercises

10:30 AM

LEAN: Proof Assistant Demo Template

- Structural Example: Definition
- Key Axioms & Lemmas
- Theorem Application Steps
- Formalization Discussion

Lean Editor.pro

Monaco

```
theorem add_assoc (n m k : N) : n + m + k = n + (m + k) :=
begin
  induction n with n' hn,
    rw add_zero,   -- Base Case: 0 + m + k = 0 + (m + k)
    rw zero_add,
    rw add_succ,   -- Inductive Case: succ n' + m + k = succ n' + (m + k)
    rw hn,         -- Use inductive hypothesis
    rw add_succ,   -- etc.
end
```

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TIP: Press CMD+P to Verify.

10:30 AM

Your Slide Title Here

- Your first point
- Your second point
- Your third point
- Your fourth point

filename.ext

Monaco

```
-- Replace this with your own code, pseudocode, or text.
-- The box wraps long lines automatically.
your_function(argument):
    return result
```

TIP: A short tip or callout goes here.

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10:30 AM

LEAN: Complete Proof Walkthrough

Lean Editor.pro

Monaco

```
theorem mul_comm (n m : N) : n * m = m * n :=
begin
  induction n with n' hn,
    rw zero_mul,       -- Base Case: 0 * m = m * 0
    rw mul_zero,
    rw succ_mul,       -- Inductive Case: succ n' * m = m * succ n'
    rw hn,
    rw mul_succ,
end
```

TIP: Press CMD+P to Verify.

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LEAN: Course Roadmap

- ☐ Week 1: Propositional Logic
- ☐ Week 2: Natural Numbers & Induction
- ☐ Week 3: Lists & Recursion
- ☐ Week 4: Tactics Deep Dive
- ☐ Week 5: Building a Library
- ☐ Week 6: Final Project

TIP: No code today - just planning.

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LEAN: Simplifying a Proof

Before

proof_v1.lean

Monaco

```
theorem add_zero'
  (n : N) : n + 0 = n :=
begin
  induction n with n' hn,
    refl,
    rw add_succ,
    rw hn,
end
```

After

proof_v2.lean

Monaco

```
theorem add_zero'
  (n : N) : n + 0 = n :=
by induction n;
   simp [*, add_succ]
```

TIP: The simp tactic closes both cases at once.

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A Note, Definition, or Quote

Notes.txt

Monaco

Replace this paragraph with your own text. The macbox environment gives you the same window chrome as macwindow, but typesets ordinary LaTeX – paragraphs, lists, tables, images – instead of code.

- Works with itemize, **bold**, and *italic*

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TIP: Swap this box for a table, image, or blockquote.

LEAN: Simplifying a Proof

proof_v1.lean

Monaco

```
theorem add_zero'
  (n : N) : n + 0 = n :=
begin
  induction n with n' hn,
    refl,
    rw add_succ,
    rw hn,
end
```

TIP: The simp tactic closes both cases at once.

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By the Numbers

128
theorems proved

42
lemmas reused

6
weeks of work

TIP: Swap in your own three (or more) headline numbers.

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References

Just for the citations [**texbook**][**latex:companion**][**latex2e**][**knuth:1984**]

References